Dapat laging H.A.N.D.A. (health analytics for NCR's disease anticipation): enhancing pandemic response through advanced health analytics in Metro Manila (Anna Katrina A. Ocampo)

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Abstract (Anna Katrina A. Ocampo)

The COVID-19 pandemic exposed several weaknesses of the Philippines health informatics and present opportunities for improves for a more accurate health data infrastructure, holistic analytics, and more accurate outbreak anticipation. This research proposes the "Dapat Laging H.A.N.D.A. (Health Analytics for NCR's Disease Anticipation) project," a health analytics framework designed to transform historical data into a data-driven insights for pandemic preparedness. With the use of the Department of Health's data related to the pandemic confirmed cases from 2020 – 2023, this framework combined descriptive epidemiology to establish demographic and temporal foundations through deep neural models such as LSTM, GRU, BiLSTM, Self-Attention, and BiLSTM-SA to capture nonlinear dynamics in MATLAB R2025b. The study revealed three key findings: cases primarily clustered among those 20–39 years old (63.2%); the Case Fatality Rate (CFR) increased with age and was higher among males; and Pasig exhibited an outlier CFR that warrants sensitivity checks. In addition, deep learning models achieved high accuracy across outcomes: daily cases (GRU, BiLSTM-SA $R^2 \geq 0.995$), hospital admissions (BiLSTM/BiLSTM-SA $R^2 \approx 0.995$), deaths (GRU $R^2 = 0.998$), and recoveries (LSTM $R^2 = 0.995$). The framework employs a data-centric methodology that maintains high scalability. Deep learning offers accuracy. It maintains interpretability. This strengthens the National Capital Region (NCR). The framework improves real-time surveillance. It increases healthcare preparedness. It builds workforce resilience for health crises.

Keywords: health analytics, deep neural learning, forecasting, COVID-19, pandemic preparedness (Anna Katrina A. Ocampo)

1. **Introduction** (Anna Katrina A. Ocampo)

The 2020 global pandemic caused by Covid-19, changed the world's economies, social life and especially the public health systems by scientific discoveries such as quick distribution of vaccines that reduce the severity of the transmission of the disease (Salman, n.d.; Tyagi & Sharma, 2024), causing the global recession since the 1930s extensive labor market and financial disturbance (S. Kumar et al., 2025), and transforming social interaction, work routines, and community adaptability (Joshi, 2023). Due to this phenomenon, healthcare systems were pushed to their limits, also leading to unstable workforces and economic disruptions. According to (Haw et al., 2020), the Philippines' National Capital Region (Metro Manila) became the epicenter, accounting for 68.8% of cases where high population density accelerated spread and necessitated stringent public health measures. High density and socio-economic disparities made public health measures harder to implement and sped up the spread, especially in cities like Manila and Quezon City where crowded areas made it harder to keep distance (Salva Villarama et al., 2021; Vicente E. Montaño, 2022).

As the Philippines' primary economic and commercial hub, Metro Manila's huge number of businesses, transport networks, and workplaces drove high mobility that heightened COVID-19 exposure for the working-age population (18–59), who comprised the majority of confirmed cases and bore the brunt of employment losses, particularly in transportation-related sectors like retail, services, and manufacturing; severe and prolonged lockdowns precipitated unforeseen instability as urban mobility fell to 26% of pre-lockdown levels before recovering to 70% by August–September 2020, with smaller declines in cities more oriented to work-from-home sectors (Jiang et al., 2022), while the National Capital Region suffered minimum cumulative losses equal to 10.66% of GRDP (de Lara-Tuprio et al., 2022) and MSMEs faced constant drops in demand and revenue that persisted into the recovery (Shinozaki & Rao, 2020); the crisis also had gendered labor effects, with women especially those with minor children experiencing greater employment setbacks due to increased care burdens and distance learning (Ducanes & Ramos, 2023) and a rise in informal work amid limited welfare support (Ramos, 2024); adding to these pressures, the prolonged suspension of public transport among the world's longest severely constrained access to essential services, including healthcare, undermining well-being for transit-dependent residents (Sunio et al., 2023). Google Mobility Reports showed that during Enhance Community Quarantine (ECQ), workplace and transit-station activity dropped by up to 80%, showing that people were rapidly shifting to remote work and using less public transportation in order comply with stay-at-home orders (Sadowski et al., 2021). This was necessary for public safety, but it caused significant productivity and income losses for wage-dependent workers in the NCR's commercial districts.

In the face of these difficulties, health analytics combining data science, statistical modeling, and AI turned pandemic data into useful information for tracking the spread, assessing interventions, and predicting outbreaks. This type of information is essential for optimizing testing, preparing for surges, and distributing resources. However, health analytics often stayed fragmented and retrospective and failed to integrate well with socio-economic and business data that affected how the pandemic spread. For example, AI and machine learning models helped monitor and forecast diseases by understanding transmission patterns and how policies would affect decision-makers (Butt, 2024; Trump et al., 2025).

With this, the researchers aimed to develop a framework for a comprehensive health analytics initiative titled "Dapat Laging H.A.N.D.A. (Health Analytics for NCR's Disease Anticipation) Project," which seeks to integrate demographic and epidemiological data from public health stakeholders into a data-driven system that enhances current predictive analytics for real-time, data-driven decision-making. The goal of the HANDA project is to improve workforce resilience, business continuity, and the overall adaptive capacity of working-age Filipinos in Metro Manila. This segment was chosen because they are considered the group most affected by health and economic disruptions. The project leverages experiences from the COVID-19 pandemic to develop a proactive analytical health ecosystem—designed not just as a response, but as a proactive strategy in health crisis management—to make the National Capital Region of the Philippines more efficient in utilizing health informatics data.

The purpose behind the HANDA project is driven by a number of research gaps, including the lack of utilization of complete pandemic datasets in history, which present trends, patterns, and correlations across different segments for predictive modeling using other relevant descriptive statistics. In addition, analytics for pandemic surveillance in the Philippines rely on traditional statistical and machine learning applications, where the application and consideration of deeper neural architectures like LSTMs and CNNs have been very limited to date, due to limitations in short time series forecasting (Miller et al., 2023, 2024), the need for domain knowledge integration via data visualization and domain-tuned large language models (Miller et al., 2023, 2024), and ongoing interpretability issues addressed through mechanisms and hybrid designs (Anshul et al., 2024; Neeraj et al., 2020); Thus, a thorough integration of strong time series forecasting into health monitoring systems reduces the anticipation of case surges, hospital demand, and resource needs, hence calling for a comprehensive Health Analytics Framework built on descriptive analytics, deep neural learning, and integrated forecasting to boost preparedness and resilience for future pandemics.

The proposed project aims to apply descriptive and inferential statistics using historical pandemic datasets of the NCR from 2020–2024. The demographic segments of the study include the top five cities of Metro Manila that were severely affected by the disease, with people affected ranging from 18 to 59 years old, to generate baseline insights on demographic, geographic, and clinical trends. Furthermore, the researchers develop, train, and validate deep neural learning models (LSTM, CNN) to capture and predict key pandemic outcomes with improved accuracy. Lastly, the outputs from descriptive and inferential statistics, deep neural learning, and forecasting will be integrated into a unified analytics framework that provides data-driven evidence for proactive public health response and resource allocation management.

This research contributes to the overall health informatics of the country as part of pandemic readiness by providing a comprehensive, data-driven health analytics framework that combines descriptive statistics, inferential statistics, deep neural learning, and forecasting to address gaps in situational awareness and predictive accuracy. Early warnings to manage hospital capacity, supplies and staffing are some of the benefits of the project that can help hospitals, healthcare providers for future pandemics. Local governments and policymakers can adjust community measures and national agencies like the DOH can align policies and funding with projected needs. Aside from the health sectors, businesses, researchers, and data scientist gain insights for analysis

and planning. General public especially cities with the most population benefits from earlier interventions, improved access to care, and reduced disruption during future outbreaks.

2. **Methods** (Anna Katrina A. Ocampo)

2.1 Research Design

Using DOH's surveillance data from March 2020 to 2023 of December which covers the City of Manila, Makati, Taguig, Pasig and Quezon and with the working-age population (18-59) as the main focus of the study, this study uses an observational, time-series analytics design to develop a health analytics framework for future pandemic preparedness in the Philippines' Metro Manila. The framework integrates descriptive epidemiology to establish baselines and heterogeneity, inferential statistics to quantify risk signals without claiming causality, and predictive modeling to generate actionable projections. Descriptive epidemiology is crucial for understanding the baseline and heterogeneity of pandemic data, which helps in identifying patterns and trends in disease spread (Bansilan & Rabajante, 2023, 2024)

2.2 Data Collection and Study Scope

Collected data with confirmed cases among working-age adults (18-59 years) in five NCR cities. These cities were selected in DOH Covid-19 Case tracker and DOH data drop covering March 2020 to December 2023. The record includes age, sex, residence, and confirmation date with recovery, death, admission and quarantine status when available. Data processing includes organizing date formats and city labels, removing duplicate data and implausible timelines. Cases with unresolved outcomes at the time of the data freeze were flagged for sensitivity analyses that affected case fatality rate denominators.

2.3 Statistical treatment and analysis

2.3.1 Exploratory data analysis and descriptive statistics

Counts, proportions, means, standard deviations, medians, interquartile ranges, modes, skewness, and kurtosis are examples of descriptive statistics that are used to summarize clinical and demographic patterns. Trends have been visualized using temporal plots, city-by-age heatmaps, and three-month moving averages. Pearson pairwise correlations among age, sex, mortality, recovery, admission, and quarantine were calculated and interpreted cautiously given binary outcome

2.3.2 Deep Neural Learning modelling

2.3.2.1 Modeling Setup

The setup involves the preparation for missing values and outliers. This is to ensure that the data will be consistent and reliable before developing and training the DNN models. Feature

engineering filled the data gaps for the model. The models followed the standardized process of data normalization, feature selection, and model training.

Based on the results gathered, BiLSTM-SA, GRU, and LSTM gave the best results. They showed the highest accuracy across daily cases, hospital result from the predictive models. It validates the potential of using the dataset from COVID-19 as an early warning system from future health crises (S. Kumar et al., 2025). As noted by authors (Devarajan et al., 2023; Ismail et al., 2022) this modeling framework supports the wider goal of transforming COVID-19 data into predictive insights that can inform public-health and workforce planning.

2.3.2.2 Dataset preparation and feature engineering

According to Devarajan et al., 2023, feature engineering transforms raw data into a meaningful input representation. It improves the performance of deep neural networks to learn complex relationship. For COVID-19 forecasting and pandemic preparedness, it serves as a bridge between machine learning and epidemiological understanding. The models used in this study must capture temporal patterns, lag effects, and non-linear dependencies available in the pandemic dataset. However, the available information in the dataset does not clearly show the information about progression and dynamic interactions. Epidemiological metrics and time-based patterns are needed into the dataset so that the models can simulate real-world transmission cycles that affect workforce availability and healthcare demand as mentioned by (Ismail et al., 2022).

As identified in the descriptive analysis under Section 3.1, Pasig City showed outlier characteristics. It had a case fatality rate of 20.94% with the smallest sample size. These irregularities require integration of epidemiological metrics and time-based patterns into the dataset. Table 12 shows how feature engineering connects the gap for the limitations by creating additional columns to reveal trends, and correlations hidden in the data.

2.3.2.2.1 Time-based features

These features track day and month progression of COVID-19. It includes weekday, month, and days since the first reported case in the Philippines. Sine and cosine columns show the cyclic nature of time. It allows models to recognize repeating trends and fluctuations of the infection.

2.3.2.2.2 Lag and rolling features

The lag and rolling features capture historical dependencies. It reflects how past values influence current cases of disease. They provide short-term and long-term memory of case trajectories, help detect trends, peaks, and shifts in disease transmissions. These also enhance the capability to learn temporal continuity and effects of delays in the data.

2.3.2.2.3 Outcome indicators

These indicators serve as dependent variables in the forecasting models. They are essential for quantifying individual and system-level outcomes, and for predicting healthcare demand and recovery trends.

2.3.2.2.4 Outcome metrics

These metrics capture temporal and proportional relationships between outcomes. They enhance the ability of the model to reflect patient flow and recovery dynamics.

2.3.2.2.5 Epidemiological metrics

The epidemiological metrics measure severity and progression. These metrics quantify transmission rates, recovery ability, fatality proportions, and pandemic wave dynamics. They also enable the models to understand the impact of the pandemic on population levels.

Table 1
Featured engineering columns added to the dataset of DOH.

Category	Featured Engineering Columns
Time-based Category	Month, WEEKDAY, IsWeekend, DaysSinceFirstCase, MonthSin, MonthCos, DayOfWeekSin, DayOfWeekCos, MonthProgress
Lag and Rolling Features	Cases_t1, Cases_t7, Cases_t14, Rolling7_Avg, Rolling14_Avg, RollingDeaths7, RollingRecoveries7, ActiveCases_t7, ActiveCases_t14, RollingActive7, GrowthRate, GrowthAcceleration
Outcome Indicators	DailyCases, IsAdmitted, IsDeceased, IsRecovered,
Outcome Metrics	RecoveryTime_Days, HospitalToRecoveryRatio, MeanRecoveryTime, AdmissionToDeathLag
Epidemiological Metrics	ActiveCases, CaseIncreaseRate, SevereCaseRatio, FatalityRate, RecoveryRate, FatalityRateChange, RecoveryRateChange, HealthStatusIndex, PandemicWaveIndex

2.3.2.3 Deep Neural Network models

The models used and implemented in MATLAB are presented below. Each model converts the input feature matrix X_t, into output $\hat{Y}_t$. The models follow the standardized format for LSTM, BiLSTM, GRU, Self-Attention, and BiLSTM–Self-Attention.

LSTM is computed based on the following formulas:

$$f_{\mathrm{t}} = \sigma(b_f + W_f[h_{\mathrm{t-1}}, X_{\mathrm{t}}]) \quad (1)$$

$$i_t = \sigma(b_i + W_i[h_{t-1}, X_t]) \quad (2)$$
$$\tilde{C}_t = tanh(b_c + W_c[h_{t-1}, X_t]) \quad (3)$$
$$C_t = i_t \odot \tilde{C}_t + f_t \odot C_{t-1} \quad (4)$$
$$o_t = \sigma(b_o + W_o[h_{t-1}, X_t]) \quad (5)$$
$$h_t = o_t \odot tanh(C_t) \quad (6)$$
$$\hat{Y}_t = W_y h_t + b_y \quad (7)$$

The input feature vector, hidden state, and cell state at time t are X_t, h_t ,C_t respectively. f_t, o_t , and i_t are the forget, output, and input gates, respectively. The bs are the bias vectors and all Ws are the weight matrices. σ denotes the Sigmoid activation. The hyperbolic tangent is $tanh$, and $\odot$ indicates the Hadamard product, and the predicted value is $\hat{Y}_t$.

BiLSTM is computed based on the following formula:

$$\overrightarrow{h_t} = LSTM(X_t, \overrightarrow{h_{t-1}}) \quad (8)$$
$$\overleftarrow{h_t} = LSTM(X_t, \overleftarrow{h_{t+1}}) \quad (9)$$
$$h_t = [\overrightarrow{h_t}; \overleftarrow{h_t}] \quad (10)$$
$$\hat{Y}_t = b_y + W_y h_t \quad (11)$$

Forward and backward hidden states are indicated by $\overrightarrow{h_t}, \overleftarrow{h_t}$. The concatenation of forward and backward state is denoted by $[.;.]$, while b_y , W_y are the bias and output weight. The other parameters are the same with LSTM.

The GRU models are computed as using the formulas:

$$z_t = \sigma(b_z + W_z[h_{t-1}, X_t]) \quad (12)$$
$$r_t = \sigma(b_r + W_r[h_{t-1}, X_t]) \quad (13)$$
$$\hat{h}_t = tanh(b_h + W_h[r_t \odot h_{t-1}, X_t]) \quad (14)$$
$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \hat{h}_t \quad (15)$$
$$\hat{Y}_t = W_y h_t + b_y \quad (16)$$

The update gate, and reset gate are z_t, and r_t respectively. The current hidden state is h_t and $\hat{h}_t$ is the candidate hidden state. The weight matrices are the Ws, and the bias vectors are the bs.

The formulas of Self-Attention (SA) models are the following:

$$Q = X_t W_Q, \quad V = X_t W_V, \quad K = X_t W_K, \quad (17)$$
$$Attention(Q, K, V) = softmax\left(\frac{(QK^T)}{\sqrt{d_k}}\right)V \quad (18)$$

$$\hat{Y}_t = W_y \, Attention(Q, K, V) + b_y \tag{19}$$

X_t is the input feature matrix, while d_k is the key dimension and $softmax$ is the activation for attention weighting. $Q, V, K,$ are the Query, Value, and Key matrices, respectively. Projection weight matrices are denoted by W_Q, W_K, W_V, while W_y , b_y are the output weight and bias.

BiLSTM-SA is a combination of BiLSTM and Self-Attention equations. It uses (8), (9), (10), and the following formulas:

$$Q = H_t W_Q, \quad K = H_t W_K, \quad V = H_t W_V \tag{20}$$

$$Attention(H_t) = softmax\left(\frac{(QK^T)}{\sqrt{d_k}}\right)V \tag{21}$$

$$\hat{Y}_t = W_y \, Attention(H_t) + b_y \tag{22}$$

The combined hidden representation from forward and backward LSTM is represented by H_t, while the other parameters are defined under Self-Attention.

The Coefficient of Determination (R^2), Mean Squared Error (MSE, Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) were computed using the following equations:

$$R^2 = 1 - (\Sigma(Y_t - \hat{Y}_t)^2) \,/\, (\Sigma(Y_t - \bar{Y})^2) \tag{23}$$

$$MSE = (1/n)\, \Sigma(Y_t - \hat{Y}_t)^2 \tag{24}$$

$$RMSE = \sqrt{MSE} \tag{25}$$

$$MAE = (1/n)\, \Sigma|Y_t - \hat{Y}_t| \tag{26}$$

$$\bar{Y} = (1/n)\, \Sigma(Y_t) \tag{27}$$

The mean of all observations is $\bar{Y}$, n is the total number of observations, and Y_t is the actual observed value at time t.

3. Results

3.1 Exploratory data analysis and descriptive statistics (Rhea Emelia B. Mergullas)

The presence of COVID-19 cases was evident across NCR and this section provides the descriptive statistics and exploratory data analyses of COVID-19 data across five metro areas in the Philippines: Manila, Quezon City, Makati, Taguig and Pasig. The data collected was used to establish the foundation for the framework. The study combines data from five NCR cities with

records, which include demographic variables such as age, sex, residence and health outcome status (recovered or died) with a total of 126,222 confirmed cases.

The dataflow for this study was summarized in Table 2. The analysis comprises of working age group from 18-59 years old with a specific observation made on core workforce (ages 25-44 years) as this age group represents the peak employment age and holds the highest number within these cities.

Table 2
COVID -19 overall dataset summary.

Characteristics	Value
Total Cases	126,222
Study Period	March 2020 - December 2023
Number of Cities	5
Mean Age (SD), years	35.78 (10.86)
Median Age, years	34
Female (%)	50.24
Male (%)	49.76
Overall CFR (%)	1.90
Hospital Admission Rate (%)	1.99
Quarantine Rate (%)	12.04

3.1.1 Age distributions

The selected age profile of the COVID cases in this study focuses on the working age population. Table 3 presents detailed descriptive statistics for the age variable with the most

frequent age of 26 years old, with the interquartile range spanning from 27 to 44 years, which aligns closely with the core workforce age which represents peak employment age.

Table 3
COVID -19 age distribution statistics.

Descriptive Statistics	Value
Count	126,222
Mean	35.78
Standard Deviation	10.86
Minimum	18
25th Percentile	27
Median (50th Percentile)	34
75th Percentile	44
Maximum	59
Mode	26
Range	41
Interquartile Range (IQR)	17
Skewness	0.46
Kurtosis	-0.86

Figure 1 shows that the age distributions were in right-skewed pattern. With the median age of 34 years old and the mean age of 35.8 years, it was confirmed that the largest cluster of cases peaks around 27 years old based on the KDE curve. The gradual decrease after the peak reflects the spread as well in the older age with fewer cases recorded. The alignment of IQR highlights that half of the cases fall within the core workforce age group.

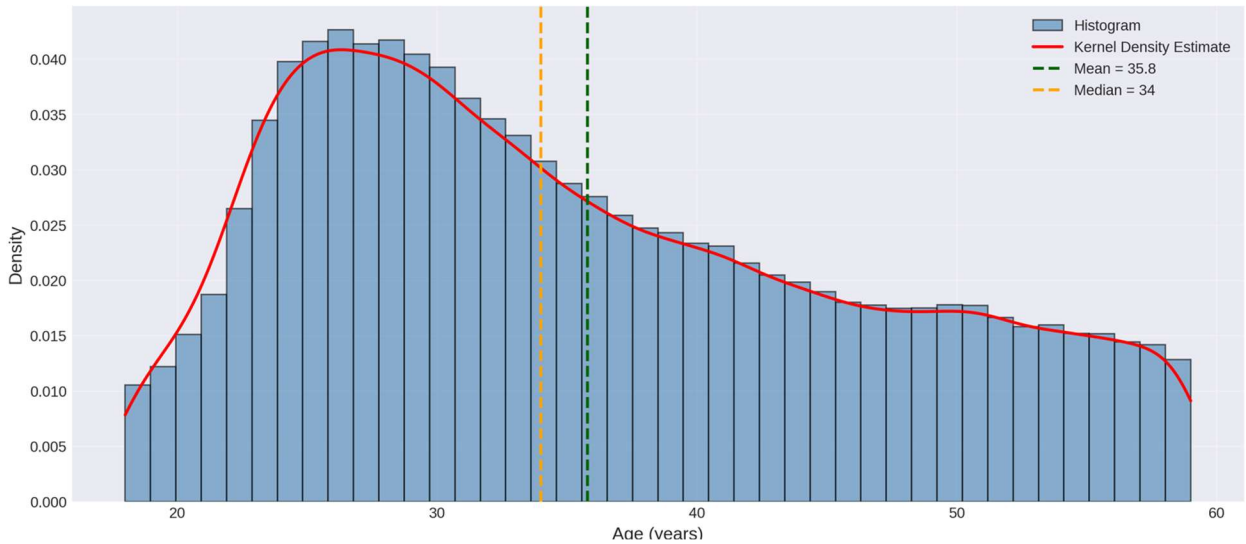

Fig. 1. Distribution of COVID-19 cases by age with histogram and kernel density estimate.

The individuals aged 25–29 years old had the highest number of cases recorded (20.29%, 25,608 cases), followed by those aged 30-34 years old (17.00%, n = 21,462) and those aged 20-24 years (13.14%, n = 16,581). The 20-39 age range represents about 63.2% of all recorded cases.

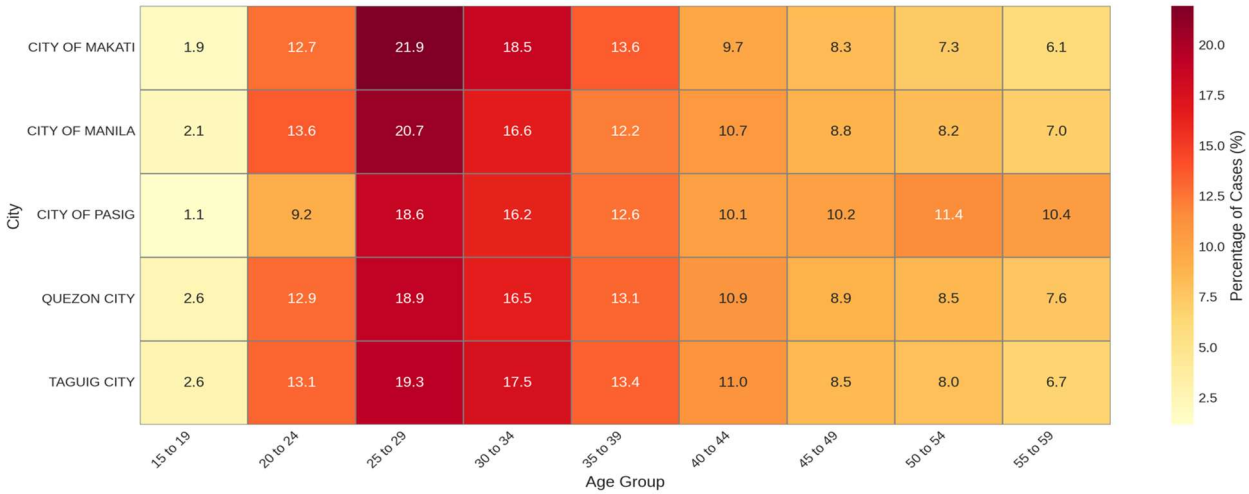

Fig. 2. Percentage of COVID-19 cases by city and age group.

As indicated in Figure 2, heatmap analysis shows the city and age group breakdown of COVID-19 cases. Collectively, the results reveal a consistent trend. Most of the reported cases from the dataset were concentrated within the younger to mid-adult age groups, although the spread differed across the five NCR cities. In Makati, the 25-29 age group has the largest reported number of cases, which is around 21.9%. By contrast, Pasig City displayed a broader spread of cases with a notable share among older working-age groups, particularly those from 50 to 54 years old. These variations likely stem from concentrated working industries in each city and commuting patterns that connect business districts.

Statistics by city level highlight differences in the average age of recorded cases. Pasig had the highest mean age, which is 38.08 years old with the standard deviation of 11.32, while Makati had

the lowest which is 35.26 years and standard deviation of 10.45. Each city showed a similar pattern in its age distribution, marked by positive skewness and a slight negative in kurtosis.

Table 4
COVID-19 age statistics per city.

City	Count	Mean	Median	Mode	Std Dev	Min	25th	75th	Max	Skewness	Kurtosis
City Of Makati	23681	35.26	33	27	10.45	18	27	43	59	0.56	-0.68
City Of Manila	54230	35.73	33	26	10.9	18	27	44	59	0.46	-0.88
City Of Pasig	1748	38.08	36	26	11.32	18	28	48	59	0.27	-1.14
Quezon City	37693	36.11	34	28	11.01	18	27	45	59	0.4	-0.91
Taguig City	8870	35.66	34	27	10.76	18	27	43	59	0.45	-0.82

3.1.2 Sex distributions

The overall sex distribution was nearly equal. Out of the total cases, 50.24% were females and 49.76% were males. Minor variations were observed in the city-level distribution: female representation varied across cities. Table 5 presents detailed gender distribution by city-level.

Table 5
Sex distribution by city.

City	Female		Male		Total
	Count	%	Count	%	
City Of Makati	12075	50.99	11606	49.01	23681
City Of Manila	26733	49.3	27497	50.7	54230
City Of Pasig	842	48.17	906	51.83	1748
Quezon City	19452	51.61	18241	48.39	37693
Taguig City	4313	48.62	4557	51.38	8870

3.1.3 Geographic distribution and city-level patterns

Manila recorded the highest case count (n = 54,230, 43.0% of total), followed by Quezon City (n = 37,693, 29.9%), Makati (n = 23,681, 18.8%), Taguig (n = 8,870, 7.0%), and Pasig (n =

1,748, 1.4%). Figure 3 displays total cases per city.

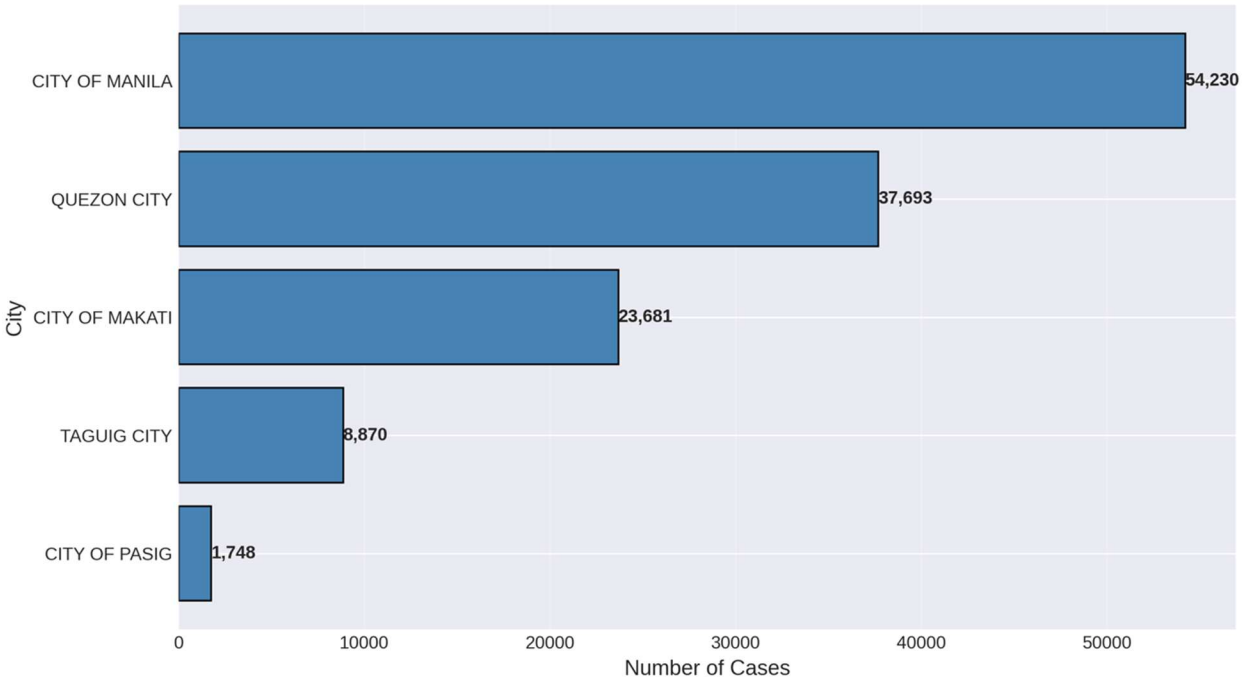

Fig. 3. Total recorded COVID cases per city.

Cities were categorized into high-intensity business districts (Makati and Taguig) and medium-intensity districts (Manila, Quezon City, and Pasig) based on the density of corporate offices. High intensity districts accounted for 25.79% of cases (n = 32,551), while moderate intensity districts accounted for 74.21% (n = 93,671). Table 6 shows the cross-tabulation of cities by age group.

Table 6

COVID -19 age group per city.

City	18 to 19	20 to 24	25 to 29	30 to 34	35 to 39	40 to 44	45 to 49	50 to 54	55 to 59
Makati City	442	3006	5186	4390	3222	2299	1965	1737	1434
Manila	1141	7388	11247	9009	6595	5793	4793	4472	3792
Pasig City	20	161	326	284	220	177	178	200	182
Quezon City	967	4861	7137	6230	4946	4107	3367	3220	2858
Taguig City	232	1165	1712	1549	1187	973	750	712	590

3.1.4 Health outcomes distributions

The overall case fatality rate was 1.9% (n = 2,399 deaths) and the recovery rate was 98.1% (n = 126,222). The following subsections provide health outcomes by age, sex and city. Age-

specific mortality analysis showed that CFR increased progressively with age. The CFR in the 18-19 age group was 0.39%, which increased to 7.46% in the 55-59 age group. Table 7 presents complete details of health outcomes by age group. In the main workforce (25-44 years), 769 deaths were recorded, which is 32.05% of the total mortality.

Table 7
Health outcomes per age group.

Age Group	Total Cases	Died	Recovered	CFR (%)	Recovery Rate (%)
18 to 19	2802	11	2791	0.39	99.61
20 to 24	16581	61	16520	0.37	99.63
25 to 29	25608	140	25468	0.55	99.45
30 to 34	21462	149	21313	0.69	99.31
35 to 39	16170	202	15968	1.25	98.75
40 to 44	13349	278	13071	2.08	97.92
45 to 49	11053	346	10707	3.13	96.87
50 to 54	10341	551	9790	5.33	94.67
55 to 59	8856	661	8195	7.46	92.54

Male had a higher case fatality rate (CFR = 2.38%) than female (CFR = 1.42%), indicating a difference of 0.96 percentage. The recovery rate for male was 97.62% and for female its 98.58%. Figure 4 displays the difference of the outcome rates from case fatality rate to recovery rate.

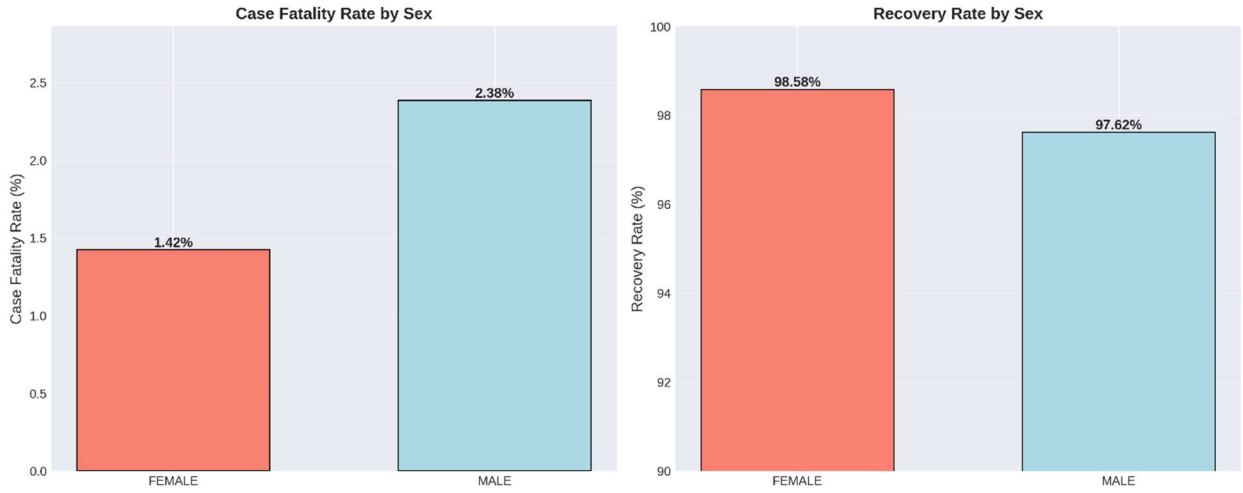

Fig. 4. Health outcome rates by sex.

City-level fatality rates varied considerably. Some city results need closer review. Pasig shows a case fatality rate of 20.94 percent, far higher than the other four cities, which range from 1.02%

to 2.69 %. which makes Pasig 7.8 to 20.5 times higher. Overall results were checked with and without Pasig. With Pasig, the case fatality rate is 1.90 percent. Without Pasig, the rate is 1.61 percent. The difference is 0.29 percentage points. Pasig changes the way city comparisons look. With only 1.4 percent of total sample cases, the effect on the total dataset is limited. Across age, sex and in correlations, the patterns stay consistent whether Pasig was included or not.

Table 8
Health outcomes per city.

City	Total Cases	Died	Recovered	CFR (%)	Recovery Rate (%)
City Of Makati	23681	242	23439	1.02	98.98
City Of Manila	54230	627	53603	1.16	98.84
City Of Pasig	1748	366	1382	20.94	79.06
Quezon City	37693	925	36768	2.45	97.55
Taguig City	8870	239	8631	2.69	97.31

3.1.5 Pearson pairwise correlation matrix

Analysis was conducted to test the connection between age, sex, and health outcomes using Pearson correlation analysis which shows in Figure 5 for the matrix and Table 9 for numerical values.

Age comparison with fatality has a weak positive association, which means older working age groups ($r = 0.140$, $p < 0.001$) had higher risk in mortality. A weak positive relationship also exists between hospital admission and mortality ($r = 0.190$, $p < 0.001$). Hence, there is a slightly higher risk of death among hospitalized patients. There is a weak relationship between gender and mortality ($r = 0.035$, $p = 0.001$). The death rate among males (coded as 1) is slightly higher as compared to females (coded as 0).

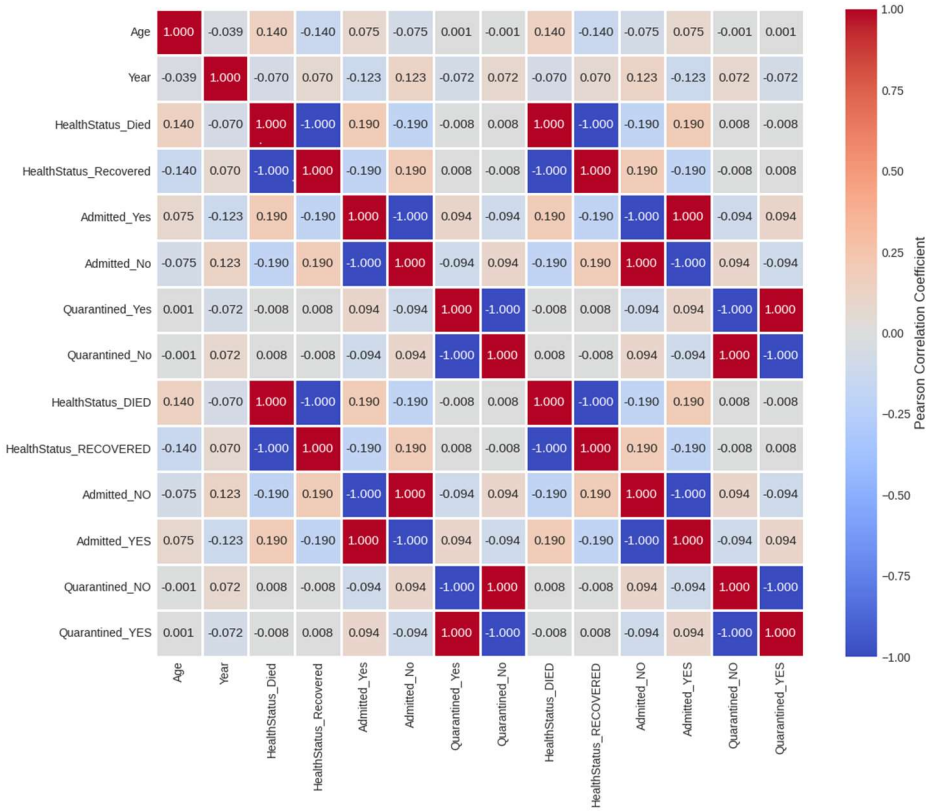

Fig. 5. Pearson pairwise correlation matrix.

The comparison of quarantine and mortality are shown to have a negative correlation (r = -0.008, p = -0.001), but it does not have a direct effect on the mortality rates. The strongest correlation calculated was between hospital admission and mortality (r = 0.190); however, the correlation value is also weak. In turn, these factors can influence mortality rates due to a vast number of factors that this dataset cannot observe.

Table 9
Pearson correlation matrix table.

	Age	Died	Recovered	Admitted	Not Admitted	Quarantined	Not Quarantined	Sex (Male=1)
Age	1	0.14	-0.14	0.075	-0.075	0.001	-0.001	0.03
Died	0.14	1	-1	0.19	-0.19	-0.008	0.008	0.035
Recovered	-0.14	-1	1	-0.19	0.19	0.008	-0.008	-0.035
Admitted	0.075	0.19	-0.19	1	-1	0.094	-0.094	0.006
Not Admitted	-0.075	-0.19	0.19	-1	1	-0.094	0.094	-0.006
Quarantined	0.001	-0.008	0.008	0.094	-0.094	1	-1	0.002
Not Quarantined	-0.001	0.008	-0.008	-0.094	0.094	-1	1	-0.002
Sex (Male=1)	0.03	0.035	-0.035	0.006	-0.006	0.002	-0.002	1

3.1.6 Temporal Distributions

Annual recorded cases had a substantial variation across the period. Cases had increased by 243% between 2020 and 2021 (from 21,849 to 74,864 cases) but then declined to 29,169 in 2022 (61% of 2021) and as low as 340 cases in 2023 (99% of 2022). The annual trend is shown in Figure 6.

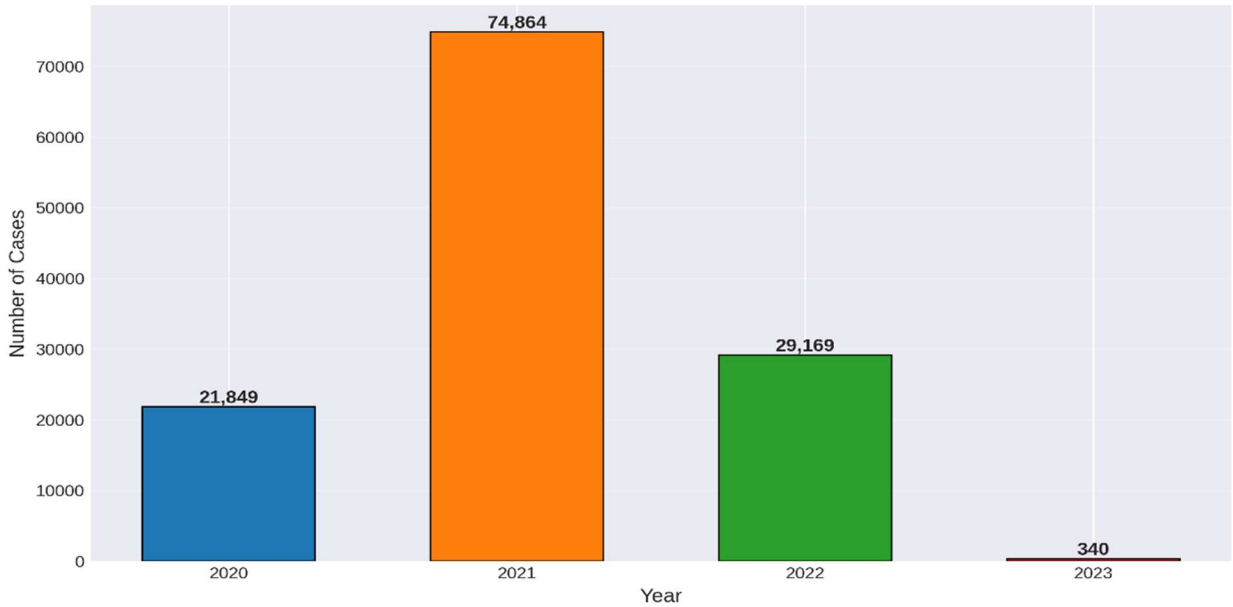

Fig. 6. Yearly COVID-19 cases trend.

Quarterly analysis showed that there are two major peaks: Q1 2021 (n=20,633 cases) and Q2 2021 (n= 28,893 cases). The reported COVID-19 cases of Q3 2020 were also high (n = 10,937). The five cities showed consistent time patterns where they had peaks in the same quarters. Figure

7 shows the trends of the quarters over the course of the study.

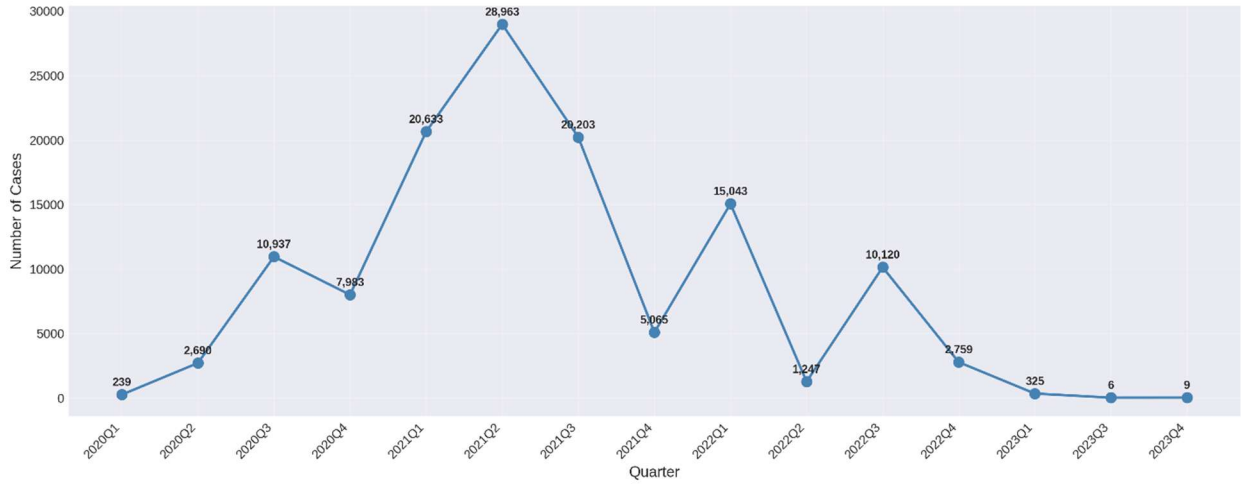

Fig. 7. Quarterly COVID-19 cases trend (2020-2023).

Monthly COVID-19 recorded cases exhibited higher month-to-month fluctuations compared to quarterly analysis, with the highest number of cases recorded by April 2021 (n=17,423 cases). The five peak months of the period are shown in Table 10. To reduce short-term variation, a three-month moving average was used to see short fluctuations and observed a long-term increase in the first half of 2021. Figure 8 shows the monthly trend with a 3-month moving average.

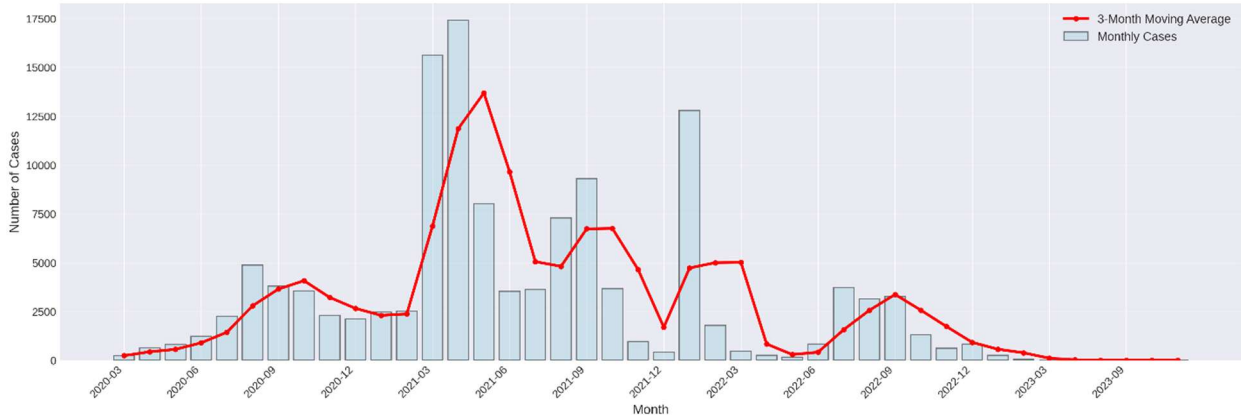

Fig. 8. Monthly COVID-19 cases with 3-month moving average (2020-2023).

Table 10
Peak months for COVID-19 cases.

Peak Month	Cases	3-Month Moving Average
2021-04	17423	11858
2021-03	15640	6878
2022-01	12793	4729
2021-09	9299	6734
2021-05	8009	13691

3.1.7 Healthcare service distributions

There was a variation in hospital admissions in different cities, which was 13.39% in Pasig, 1.27% in Taguig, 2.39% in Quezon City, 1.77% in Manila, and 1.27% in Makati. Overall, there were 2,511 cases (1.99%), in total, that were hospitalized.

Table 11
Hospital admission and quarantine status by city.

City	Total Cases	Admitted	Quarantined	Admission Rate (%)	Quarantine Rate (%)
City Of Makati	23681	301	2275	1.27	9.61
City Of Manila	54230	962	9520	1.77	17.55
City Of Pasig	1748	234	412	13.39	23.57
Quezon City	37693	901	2148	2.39	5.7
Taguig City	8870	113	848	1.27	9.56

Quarantine rates also showed city-level variation: Manila (17.55%), Makati (9.61%), Taguig (9.56%), Quezon City (5.70%), and Pasig (23.57%). In all cities, 15,204 cases (12.04%) and 111,019 cases (87.90%), respectively, were quarantined and not quarantined. Figure 9 demonstrates city quarantine status. Table 10 shows the detailed analysis of hospitalization rates and quarantine.

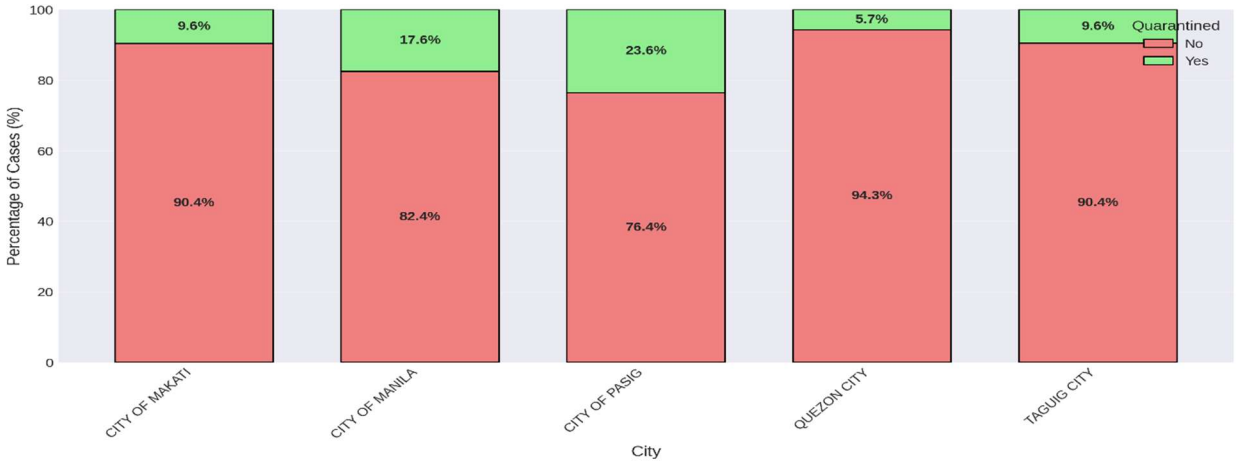

Fig. 9. Quarantine status per city.

3.1.8 Business districts by workforce analysis

Metro Manila's five target cities exhibit varying business districts intensities that shape both transmission patterns and impact in the economy. For analytical purposes, cities were classified into two categories, based on the concentration of corporate headquarters, financial institutions and commercial offices.

High Intensity Business Districts characterized by dense clustering of multinational corporations, BPOs centers, and financial institutions with high daily commuters. Medium Intensity Business Districts are blended residential, commercial and government offices with more diverse business activities.

High-intensity business districts (Makati and Taguig) had 32,551 total cases, while medium-intensity districts (Manila, Quezon City, and Pasig) had 93,671 cases. Mean ages were similar between district types: high-intensity districts averaged 35.69 years (SD = 10.84) and medium-intensity districts averaged 35.82 years (SD = 10.87). Figure 10 displays the overall case distribution by business district classification.

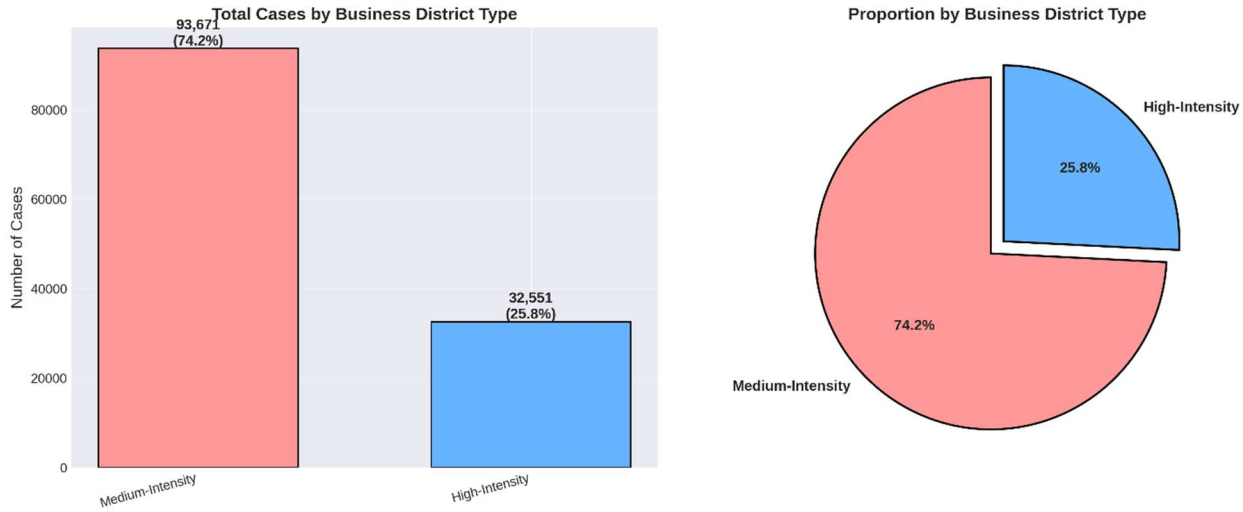

Fig. 10. COVID-19 cases distribution by business district type.

Within the working age population (18-59 years), the core workforce (age group 25 to 44 years) in cities accounts for 57.61% to 63.75% of cases. This age group represents the peak years of employment in the business districts of Metro Manila. Makati had the most cases (63.75%) that were reported. Next in line were Taguig (61.1%), Manila (60.20%), Quezon City (59.48%), and Pasig (57.61%). The detailed workforce impact by city is shown in Table 12.

The fatality rate for people aged 25 to 44 in Makati was 0.48%, and in Taguig, it was 1.42%. Separately, Pasig displayed an outlier of 13.51%. Regardless of district classification, the first and second quarters of 2021 experience short rises in cases in all five cities.

Table 12
Core workforce impact by city (ages 25-44).

City	Total Cases	Workforce Cases (25-44)	Workforce %	Workforce Deaths	Workforce CFR (%)
City Of Makati	23681	15097	63.75	72	0.48
City Of Manila	54230	32644	60.2	200	0.61
City Of Pasig	1748	1007	57.61	136	13.51
Quezon City	37693	22420	59.48	284	1.27
Taguig City	8870	5421	61.12	77	1.42

Overall, the 126,222 confirmed cases from five NCR cities (March 2020–December 2023) that serve as the basis for the framework's components showed parallel trajectories in this analysis.

3.2 **Deep Neural Learning Models** (Marita Grace G. Moog)

This section discusses the implementation and evaluation of Deep Neural Learning models in MATLAB R2025b using the preprocessed and feature engineered columns dataset as mentioned in Section 2.

The study applied Deep Neural Network (DNN) methods. These include Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), Bidirectional Long Short-Term Memory (BiLSTM), Self-Attention with Bidirectional Long Short-Term Memory (BiLSTM-SA), and Self-Attention (SA). The models were used to forecast four critical pandemic factors. These are the daily cases, hospital admissions, recovery, and mortality rates.

3.2.1 Hyperparameters and model training

Each model used in the DNN was designed and optimized using hyperparameters summarized in Table 13. The setup included defined model layers, activated functions, and applied L2 regularization to improve generalization and prevent overfitting (Khan et al., 2022; Naseem et al., 2021). As mentioned by the author (S. Kumar et al., 2025), to ensure that the models will be robust, essential hyperparameters were adjusted to achieve a balance between "good fit" and "good capacity".

The models were implemented Adam optimizer and trained using over 200 to 600 epochs. The train and validation split was 70-30. Hyperparameter stayed consistent across forecasting models to ensure fair comparisons with traditional time-series models.

The performance was evaluated using Coefficient of Determination (R^2) to check the goodness of fit. The Mean Squared Error (MSE) is for the overall model error. The Root Mean Squared Error (RMSE) is used to measure the error of a model, while Mean Absolute Error (MAE) is used to compute for the difference between predicted and actual results.

Table 13
Summary of the hyperparameters used in Deep Neural Learning models.

Output Variable	Method	Hidden Units / Layers	Dropout Rate	Learning Rate	Max Epochs	Batch Size	L2	Attention Parameters
Daily Cases	LSTM	128–64	0.2	0.010	200	Default	0.0005	—
	BiLSTM	128–64	0.2	0.010	200	Default	0.0005	—
	BiLSTM-SA	128	0.3	0.010	200	Default	0.0005	Heads = 8, KeyChannels = 256
	GRU	64–32	0.3	0.005	200	Default	0.0005	—
	SA	FC 64–32	0.25	0.003	250	Default	0.0007	Heads = 4, KeyChannels = 128
Hospital Admissions	LSTM	128–64	0.1	0.001	300	64	—	—
	BiLSTM	64	0.3	0.005	150	Default	—	—
	BiLSTM-SA	64	0.3	0.005	150	Default	—	Heads = 4, KeyChannels = 128
	GRU	64	0.2	0.005	150	Default	—	—
	SA	FC 512–256–128	0.2	0.0003	600	64	0.0003	Heads = 8, KeyChannels = 256
Death Forecasting	LSTM	128–64	0.2	0.010	200	Default	0.0005	—
	BiLSTM	128–64	0.2	0.010	200	Default	0.0005	—
	BiLSTM-SA	128	0.3	0.010	200	Default	0.0005	Heads = 8, KeyChannels = 256
	GRU	64–32	0.3	0.005	200	Default	0.0005	—
	SA	FC 64–32	0.25	0.003	250	Default	0.0007	Heads = 4, KeyChannels = 128
Recovery Forecasting	LSTM	128	0.2	0.001	300	Default	—	—
	BiLSTM	128	0.3	0.001	300	Default	—	—
	BiLSTM-SA	96	0.3	0.001	300	Default	—	Heads = 4, KeyChannels = 128
	GRU	64	0.2	0.001	250	Default	—	—
	SA	FC 128–64	0.3–0.2	0.001	500	64	—	Heads = 4, KeyChannels = 128

3.2.2 Output variables result

The deep neural network models focused on four main output variables, Daily Cases, Hospital Admissions, Death, and Recoveries Forecasting. These outputs measure the spread and severity of the disease and predict future pandemic preparedness. These variables also simulate the spread of disease, effects in healthcare system, mortality and recovery of patients.

The results were evaluated by fit and capacity. A model showing a good capacity indicates that the model can generalize well beyond the training data, and a good fit means strong alignment between predicted and actual values.

3.2.2.1 Daily cases forecasting

As shown in Table 14, BiLSTM-SA (R^2 = 0.9952) and GRU (R^2 = 0.9958) achieved the highest accuracy, both showing excellent fit good capacity. Fig. 11 shows tight clustering along the line which indicates high alignment between actual and predicted values. These results support early detections of workforce exposure surges and potential absenteeism. They add value for operational planning and public health preparedness. This also contributes to supporting timely interventions during outbreaks.

Table 14

Summary of daily cases forecasting results.

Model Type	R^2	MSE	RMSE	MAE	Fit Evaluation	Capacity Evaluation
LSTM	0.9903	0.0096	0.0980	0.0667	Good	Good
BiLSTM	0.9925	0.0075	0.0865	0.0503	Good	Good
BiLSTM-SA	0.9952	0.0048	0.0692	0.0461	Excellent	Good
GRU	0.9958	0.0042	0.0646	0.0384	Excellent	Good
SA	0.9904	0.0095	0.0976	0.0813	Good	Good

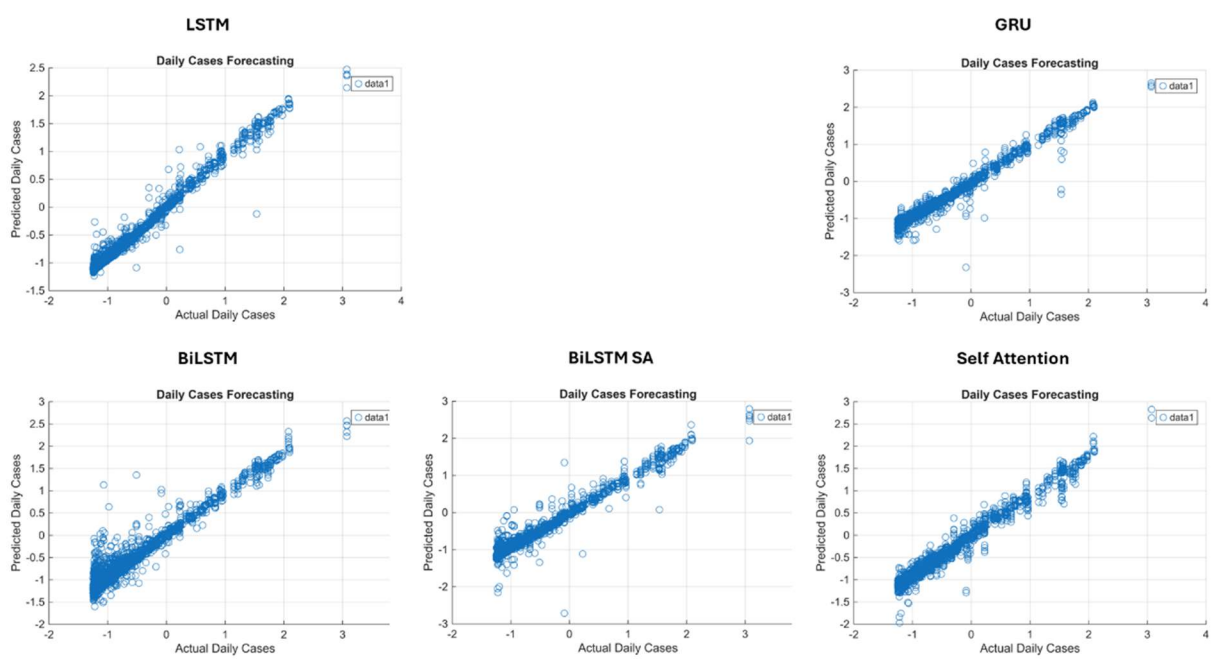

Fig. 11. The scatter plots for LSTM, GRU, BiLSTM, Self-Attention, and BiLSTM-SA models show strong correlation between predicted and actual case counts.

3.2.2.2 Hospital admissions forecasting

Table 15 shows BiLSTM-SA and BiLSTM gave the most accurate hospital admission forecasts. Both reached $R^2 \approx 0.995$. This indicates excellent fit and high predictive reliability. The scatter plots in Fig. 12 show a strong linear correlation, with minor deviations. The LSTM model stayed consistent, while GRU and SA showed underfitting. Among all models, BiLSTM-SA and BiLSTM are the most capable in hospitalization dynamics. Accurate admission forecasts can help public and private institutions plan for hospital capacity.

Table 15

Summary of hospital admissions forecasting results.

Model Type	R^2	MSE	RMSE	MAE	Fit Evaluation	Capacity Evaluation
LSTM	0.9921	0.0078	0.0884	0.0370	Good	Good
BiLSTM	0.9947	0.0053	0.0725	0.0256	Excellent	Good
BiLSTM-SA	0.9946	0.0052	0.0724	0.0241	Excellent	Good
GRU	0.9723	0.0272	0.1648	0.0325	Moderate	Slightly Under
SA	0.8635	0.1397	0.3737	0.0539	Under	Low

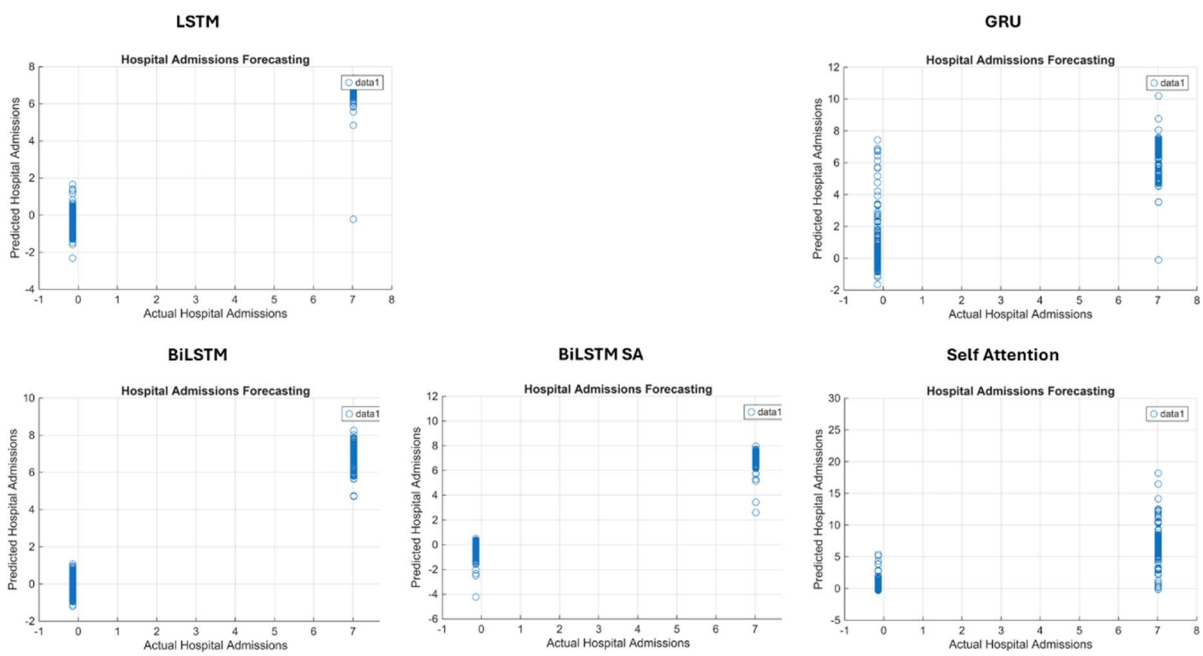

Fig. 12. The scatter plots for hospital admissions reveal a highly linear correlation, with predicted values closely matching actual hospital demand levels.

3.2.2.3 Death forecasting

Table 16 shows that the GRU model was the best model with $R^2 = 0.9977$ and with very low RMSE = 0.0495. The models, BiLSTM and LSTM, also achieved high accuracy ($R^2 \approx 0.991$). Fig. 13 shows a tight clustering near the diagonal. BiLSTM-SA and SA models underperformed which

indicates under-capacity. Overall, GRU is the ideal for mortality trend analysis. These results support public-health contingency planning for high-risk business zones.

Table 16
Summary of death forecasting results.

Model Type	R^2	MSE	RMSE	MAE	Fit Evaluation	Capacity Evaluation
LSTM	0.9910	0.0086	0.0925	0.0486	Good	Good
BiLSTM	0.9918	0.0085	0.0921	0.0704	Good	Good
BiLSTM-SA	0.9711	0.0284	0.1685	0.1157	Moderate	Slightly Under
GRU	0.9977	0.0025	0.0495	0.0121	Excellent	Good
SA	0.9582	0.0422	0.2053	0.0785	Moderate	Slightly Under

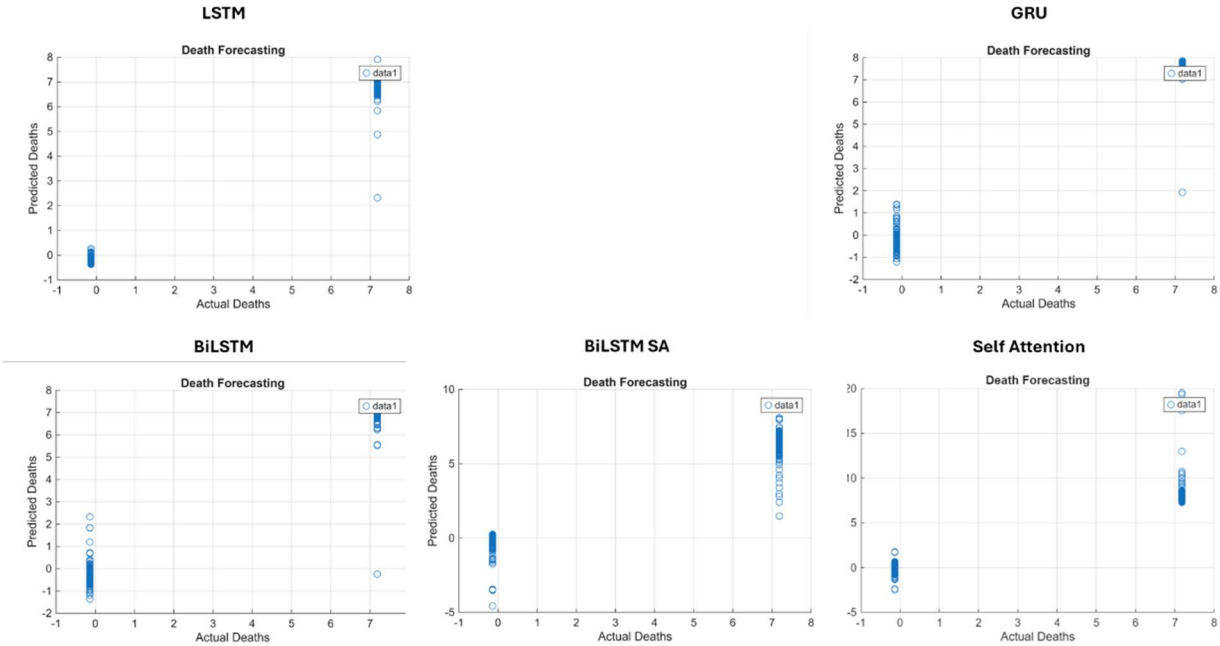

Fig. 13 The scatter plots for death forecasting show their capacity to model delayed death occurrences following infection peaks.

3.2.2.4 Recovery forecasting

Table 17 shows LSTM model ($R^2 = 0.9951$) generated the most accurate recovery forecast. GRU and BiLSTM followed with $R^2 \approx 0.9885$. The BiLSTM-SA is good fit but lower capacity for rapid recovery shifts. Fig. 14 shows a near-perfect linear relationship between actual and predicted recoveries. RMSE = 0.0681, MAE = 0.0286 confirm the good capacity and strong generalization. LSTM-based recovery models provide accurate recovery forecasts and contribute to workforce planning.

Table 17
Summary of recovery forecasting results.

Model Type	R^2	MSE	RMSE	MAE	Fit Evaluation	Capacity Evaluation
LSTM	0.9951	0.0046	0.0681	0.0286	Excellent	Good
BiLSTM	0.9885	0.0119	0.1092	0.0605	Good	Good
BiLSTM-SA	0.9827	0.0168	0.1297	0.0684	Good	Slightly Under
GRU	0.9886	0.0115	0.1074	0.0563	Good	Good
SA	0.9819	0.0189	0.1375	0.0303	Good	Slightly Under

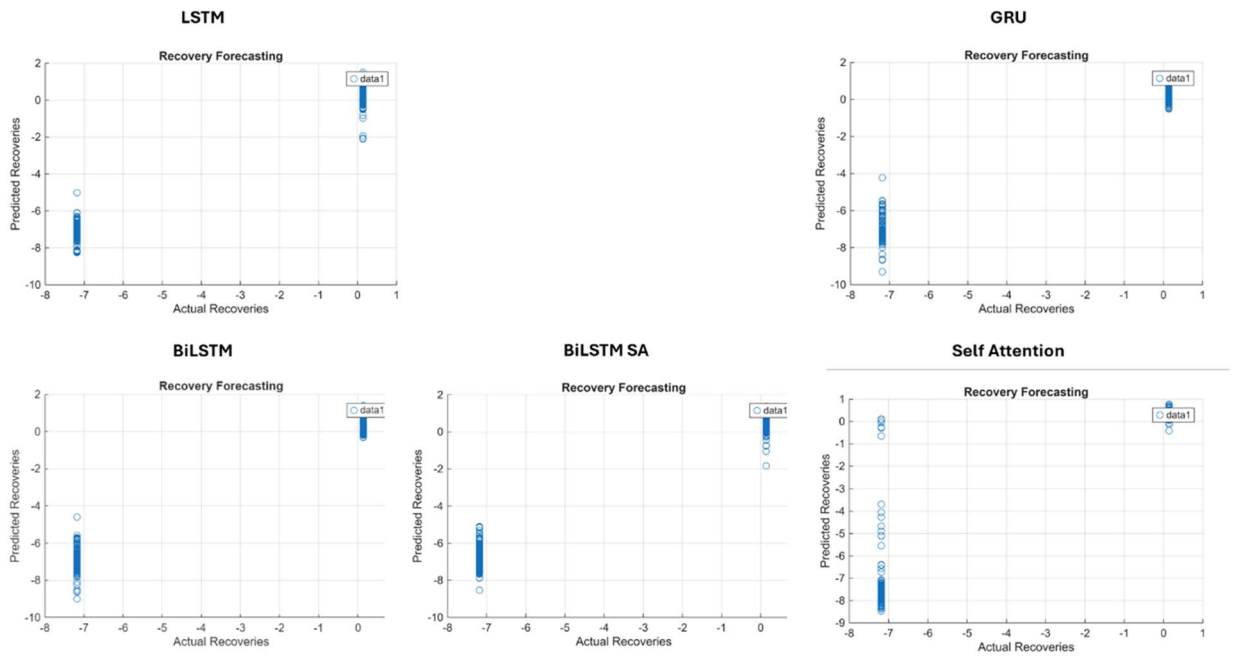

Fig. 14. The scatter plots for recovery forecasting alignment of predicted versus actual recoveries.

3.2.3 Problems addressed by the DNN forecasting results

The DNN forecasting models in MATLAB addressed the challenges outlined in the problem statement. They produced actionable insights for public health management.

The forecasting models create early warnings for outbreaks which can support timely decisions. It can also guide the capacity planning and resource allocation during pandemics. It also supports decisions for critical care and contributes to the evaluation of healthcare efficiency and patient outcome. Each model showed both good fit and good capacity, and the deep learning framework proved to be feasible for pandemic preparedness.

The integration of DNN forecasting models created a health preparedness framework. This framework provided a comprehensive, and data-driven foundation for pandemic reporting and resource management.

4. **Discussion** (Kasten Moralde)

The goal of this study was to establish an analytic framework for health that focuses on the National Capital Region, Philippines. Moving toward a data driven approach, this framework was built through distinct yet complementary analytical pillar: Descriptive Statistics and Deep Neural Learning. The results confirm that the usage of these methodologies provides an actionable insight from the pandemic which effectively addressing the identified research gaps.

4.1 Exploratory data analysis and descriptive statistics

A major limitation across many studies was the reliance on insufficient dataset often constrained by short time spans, single location focused, or inconsistent data entries. For an instance, some studies relied on datasets that covers a short timespan making it unsuitable for long term predictions (S. Kumar et al., 2025).

In this study, the historical dataset of the COVID-19 Pandemic from 2020 to 2023 from the Department of Health, Philippines, was used and transformed to provide meaningful insights using Descriptive Statistics which fulfills the first specific objective. This study focuses on the working age group that is from 18 to 59 years old from the top five business driven cities in NCR, wherein the result shows that the most affected age range with a mean of 35.78 years old within the target group was 20 to 39 years old accounting for a significant 63.2% of all the cases from Makati, Manila, Pasig, Quezon, and Taguig City. In a study conducted in 2020, the reported median was 46 years old (Haw et al., 2020). This suggests a shift in the demographic impact over time.

Additionally, there was a rise in the Case Fatality Rate (CFR) from 0.37% to 7.46% from 20-24 to 55-59 age group respectively. A positive correlation value of $r = 0.14$ was detected concerning CFR and age. This can be interpreted that age was a critical factor in the fatality rate. The finding was consistent with other studies that concludes age as well as comorbidity heightened the risk of fatality (Malundo et al., 2022; Shafigh et al., 2023). This can be a basis in vaccine prioritization for contagious diseases in the future. Also, it turned out that males displayed a higher CFR of 2.38% compared to only 1.42% for females, suggesting that gender is also a factor but might require a further clinical investigation.

Furthermore, it was also notable in the results that Pasig City had a 20.94% CFR but deeper investigation into the dataset shows that the total recorded cases in Pasig City was only 1748 cases compared to the 25244 average cases of the five cities. Despite having an insignificant impact on the overall assessment, it shows the importance of accurate data entry since data quality issue might be the culprit. Also, the synchronized pandemic peaks of the five cities from Quarter 1 to 2 of the year 2021 regardless of the location, age, and gender was a known pandemic wave identified in a similar study (Chu et al., 2022; Rampal et al., n.d.). This suggests that national health protocols, disease variant, or mobility patterns are the factors of COVID-19 transmission in the National Capital Region.

4.2 Deep neural learning models

Deep neural learning models are suggested to be used for several tasks related to COVID-19 assessment and prediction (Khan et al., 2022; R. John Martin, 2023). It is recommended to use Convolutional LSTM, Stacked LSTM, or Bi-directional LSTM for studies related to time-series

forecasting since these methods can capture long-term dependencies and temporal nonlinearities (Arshad & Algarni, 2026; S. Kumar et al., 2025).

Although when these models are being utilized, their performance is often constrained because they require a large amount of data which is frequently unavailable for training, validation, and testing. In cases where there is a data scarcity issue, a comparative analysis revealed that simpler models were sometimes powerful than complex models (S. Kumar et al., 2025)

Deep Neural Learning was applied to address the lack of advance non-linear modeling in previous studies. Using Feature Engineering, the methods LSTM, GRU, BiLSTM, SA, and BiLSTM-SA, were equipped to learn complex patterns and non-linear dependencies from the dataset of around 126k cases.

Daily Cases, Hospital Admissions, Deaths, and Recoveries were the four output variables observed via Forecasting. Every method was applied to each forecasting tasks totaling to 20 executions. All top performing models achieved an $R^2 \geq 0.99$ which demonstrates the ability of Deep neural learning models to capture complex patterns, hence, fulfilling the second specific objective.

In machine learning, the coefficient of determination, R^2, is a measure that describes how well the model explains the target variable (Arshad & Algarni, 2026). With an $R^2 = 0.9958$ and $R^2 = 0.9977$ in Daily Cases and Death Forecasting respectively, the GRU emerged as the most effective. Usage of BiLSTM-SA and BiLSTM in Hospital Admission Forecasting demonstrated an excellent performance with an $R^2 \approx 0.995$ while LSTM was optimal for Recovery Forecasting with an $R^2 = 0.9951$.

A steady outcome of "Excellent Fit" and "Good Capacity" is a notable finding that confirms the ability of these methods to predict the unseen future data which might be critical for an early warning system. The success of these methods provides the following insights: Daily Cases for early surge detection; Hospital Admissions for capacity planning; and Death/Recovery for mortality risks and healthcare system evaluation.

4.3 Integrated Framework and Future Direction

Development of a comprehensive analytics framework is the third objective this this study. Wherein identifying the context, prediction, and operation serves as the foundation of the framework. Context was determined via descriptive statistics that provides a validated answer to the question where, when, and who was affected. Meanwhile, prediction via long-term forecast was achieved using deep neural learning models that can effectively capture complex non-linear dynamics in critical variables.

4.4 Recommendations

Since the working age which was 18 to 59 years old was the focus of this study, resource allocation efforts should be prioritized for this group to reduce health risks and economic disruption. The rise of CFR in correlation with age and gender implies the need for a strategic planning and effective decision-making for these vulnerable sub-groups. Moreover, the wide variation of CFR in Pasig City, indicates the need of system-wide review for data collection and

validation to ensure that high level analytics will be based on reliable dataset. For prediction, this study recommends the development of an early warning system wherein deep neural learning will be utilized. The deep neural learning models should be used for long-term planning. These models can provide non-linear insights that can used to adjust the hospital capacity, anticipate critical care demand, and strengthen the healthcare system against outbreak.

5. Research Conclusion (Kasten Moralde)

A health analytics framework for pandemic preparedness in the National Capital Region was developed in this study by addressing research gaps. Beginning with the systematic analysis using descriptive statistics of 126,222 COVID-19 cases from the working age group of 18 to 59 years old where risks factors were identified such as increasing age and gender. This provided a demographic and epidemiological basis for strategic planning. Then deep neural learning models such as GRU, BiLSTM, and LSTM were identified to have an exceptional predictive accuracy that can be utilized for long-term forecasting considering the non-linear dynamics of Daily Cases, Hospital Admissions, Deaths, and Recoveries. The forecasts results can be used to determine the required hospital capacity and mortality. The framework effectively utilized these methods to establish a comprehensive data-driven foundation that can significantly enhance NCR's resilience against future pandemic.

6. References (Kasten Moralde)

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